Defence Institute of Advanced Technology

Deemed to be University, Girinagar, Pune



DEPARTMENT OF APPLIED MATHEMATICS

Course: Data Science Tools & Techniques

Course Code: AM609L

SIR Epidemic Model with ODE Solvers

DECEMBER 2026

Submitted by

Sanghakara Manwar	26-27-13
Rohit Satpute	26-27-14
Shreya	26-27-15
Tanesh Devhare	26-14-05

Contents

Abstract **1**

1 Introduction **1**

1.1 Motivation 1

1.2 Objectives 1

2 Mathematical Foundations **2**

2.1 The SIR model 2

2.2 The basic reproduction number 2

2.3 An analytical invariant 2

3 Numerical Methods **2**

4 Datasets and Preprocessing **3**

4.1 COVID-19 case data 3

4.2 Constructing $I(t)$ and $R(t)$ 3

5 Fitting Procedure **4**

5.1 Effective population 4

5.2 Objective function 4

5.3 Two-stage grid search 4

6 Results: Numerical Methods **4**

6.1 Analytical invariant 5

7 Results: Parameter Estimation **6**

7.1 India, second wave 6

7.2 Japan, summer wave 6

7.3 Comparison 7

8 Results: Flattening the Curve **8**

9 Historical Bombay Plague Data Application **8**

10 Discussion **9**

10.1 Choice of solver 9

10.2 Quality of the fits 9

10.3 Identifiability and boundary warnings 9

10.4 Limitations 10

11 Conclusion and Future Work **10**

References **10**

List of Figures

1 Selected waves. Left: India, daily new cases, February–August 2021. Right: Japan, daily new cases and 7-day average, July–October 2022. 3

2 Numerical solutions S, I, R from Euler and RK4 against the reference solution. At this scale the three curves are indistinguishable by eye. 4

3 Log–log convergence plot. Observed order: Euler 0.99, RK4 3.86. 5

4 Left: infected population from the numerical $S(t)$ and the analytical relation. Right: absolute deviation from the invariant, on a log scale. 6

5 India. Left: actual and predicted weekly new infections. Right: fitted S, I and R for the effective population. 6

6 Japan. Left: actual and predicted weekly new infections. Right: fitted S, I and R. 7

7 Fitted β, γ, effective fraction (left axis) and R_0 (right axis) for the two countries. 7

8 Modelled infected population under higher and lower transmission. Left: India. Right: Japan. Dashed red line: illustrative healthcare capacity. 8

9 Bombay plague, first wave (1905–06): actual and RK4-predicted weekly new cases. 9

List of Tables

1 Maximum absolute error against the reference solution. 5

2 Fitted parameters and goodness of fit. 7

3 Scenario results. R_0 in the India rows uses $\gamma = 0.019$, as in the scenario code. . . 8

Abstract

This report presents a from-scratch Python implementation of the classical SIR (Susceptible–Infected–Recovered) epidemic model and applies it to real COVID-19 data. Two explicit ODE solvers, the forward Euler method and the classical fourth-order Runge–Kutta method (RK4), are implemented and compared against a high-accuracy DOP853 reference solution. The observed orders of convergence are 0.99 for Euler and 3.86 for RK4, and RK4 preserves the analytical invariant of the SIR system to 6×10^{-8}, against 3.6 for Euler.

The transmission rate β, recovery rate γ and an effective susceptible fraction are then estimated by a two-stage grid search on weekly new-case counts, for India's second wave (February–August 2021) and Japan's summer wave (July–October 2022). India yields $\beta = 0.095$, $\gamma = 0.0196$ and $R_0 \approx 4.85$ (weekly $R^2 = 0.81$); Japan yields $\beta = 0.266$, $\gamma = 0.109$ and $R_0 \approx 2.44$ ($R^2 = 0.95$). Finally, a "flattening the curve" experiment lowers β by about 26% in both countries. This cuts the modelled peak by 30% in India (delayed by 52 days) and by 73% in Japan (delayed by 16 days). The report also discusses the limitations of the approach, notably that some fitted values sit on grid boundaries and that the fitted India infectious period is implausibly long.

1 Introduction

Compartmental models describe how an infection spreads through a population by sorting individuals into a few states and specifying the rates of movement between them. The SIR model is the simplest useful example. Its three equations have no closed-form solution in time, so they must be solved numerically, and the quality of the numerical method affects every conclusion drawn from the model.

1.1 Motivation

Epidemic forecasts and intervention studies depend on two things: a numerical solver that is accurate enough, and parameters that are consistent with observed data. This project examines both. It tests how much accuracy is lost by using a first-order method such as Euler instead of RK4, and it asks whether a three-parameter SIR model can reproduce the shape of real epidemic waves in two very different countries.

1.2 Objectives

The project covers four tasks.

- Implement the Euler and RK4 methods for the SIR equations, and verify convergence orders and the analytical invariant.
- Estimate β and γ by grid search against real COVID-19 data (India and Japan).
- Compute R_0 and run a "flattening the curve" scenario analysis.
- Bonus: fit historical Bombay City plague data (the executed analysis covers 1896–97).

Section 2 gives the model, Section 3 the numerical methods, Section 4 the data, and Section 5 the fitting procedure. Sections 6 to 8 present the results, Section 10 discusses them, and Section 11 concludes. The Bombay plague bonus is discussed in Section 9.

2 Mathematical Foundations

2.1 The SIR model

A population of size N is divided into susceptible $S(t)$, infectious $I(t)$ and recovered (removed) $R(t)$ individuals:

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \tag{1}$$

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \tag{2}$$

$$\frac{dR}{dt} = \gamma I. \tag{3}$$

Here β is the transmission rate and γ the recovery rate, both per day. The mean infectious period is $1/\gamma$. Adding the three equations gives $\dot{S} + \dot{I} + \dot{R} = 0$, so $S + I + R = N$ is conserved.

2.2 The basic reproduction number

The basic reproduction number

$$R_0 = \frac{\beta}{\gamma} \tag{4}$$

is the expected number of secondary infections produced by one infectious individual in a fully susceptible population. From Eq. (2), I grows initially if and only if $\beta S/N > \gamma$, that is $R_0 > 1$ when $S \approx N$. Interventions that reduce contact reduce β and therefore R_0.

2.3 An analytical invariant

Dividing Eq. (2) by Eq. (1) gives $dI/dS = -1 + \gamma N/(\beta S)$, which integrates to

$$S + I - \frac{\gamma N}{\beta} \ln S = C, \tag{5}$$

a constant along every trajectory. It provides an independent check on any numerical solution: the quantity on the left must stay constant, and its drift measures the solver's error. Equivalently, given $S(t)$ from a solver, the infected count can be recovered as $I = S_0 + I_0 - S + \frac{\gamma N}{\beta} \ln(S/S_0)$.

3 Numerical Methods

Write the system as $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$ with $\mathbf{y} = (S, I, R)^\top$ and step size h.

Forward Euler advances by $\mathbf{y}_{n+1} = \mathbf{y}_n + h\,\mathbf{f}(t_n, \mathbf{y}_n)$. It is first-order accurate: the global error is $O(h)$.

RK4 computes four slope evaluations,

$$\mathbf{k}_1 = \mathbf{f}(t_n, \mathbf{y}_n), \qquad \mathbf{k}_2 = \mathbf{f}(t_n + \tfrac{h}{2}, \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1),$$

$$\mathbf{k}_3 = \mathbf{f}(t_n + \tfrac{h}{2}, \mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2), \qquad \mathbf{k}_4 = \mathbf{f}(t_n + h, \mathbf{y}_n + h\mathbf{k}_3),$$

and sets $\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$. Its global error is $O(h^4)$ at four times the cost per step.

Reference solution. The exact solution is unavailable, so errors are measured against `scipy`'s DOP853 solver with `relative tolerance`$= 10^{-12}$ and `absolute tolerance`$= 10^{-14}$. The error of a method is the maximum absolute difference, over all three compartments and all grid points, from this reference.

Observed order. For step sizes h_j with errors E_j, the observed order is the slope of the least-squares line through $(\log h_j, \log E_j)$. A slope near 1 or 4 confirms first- or fourth-order behaviour.

4 Datasets and Preprocessing

4.1 COVID-19 case data

Daily data for India (2430 rows) and Japan (2473 rows) come from Our World in Data Dataset. Each file has total cases, new cases, deaths, population, reproduction rate, vaccination and stringency columns; the Japan file also has hospital and ICU counts. One wave is cut from each series:

- **India:** 1 February – 31 August 2021 (212 days), the second wave, with a peak of 414,188 daily cases on 7 May 2021.
- **Japan:** 1 July – 31 October 2022 (123 days), the summer wave, with a peak of 326,090 daily cases on 29 August 2022.

4.2 Constructing $I(t)$ and $R(t)$

The data contain no recovered counts, so every case is assumed infectious for $D = 14$ days after confirmation:

$$I(t) = \text{total}(t) - \text{total}(t - D), \qquad R(t) = \text{total}(t - D). \tag{6}$$

Only $I(t_0)$ and $R(t_0)$ are used, as initial conditions: $I_0 = 185{,}837$ and $R_0^{\text{init}} = 10{,}571{,}773$ for India, and $I_0 = 235{,}417$ and $R_0^{\text{init}} = 9{,}117{,}228$ for Japan. The fit itself is made to weekly new cases. The initial removed count is written R_0^{init} to avoid confusion with the reproduction number R_0.

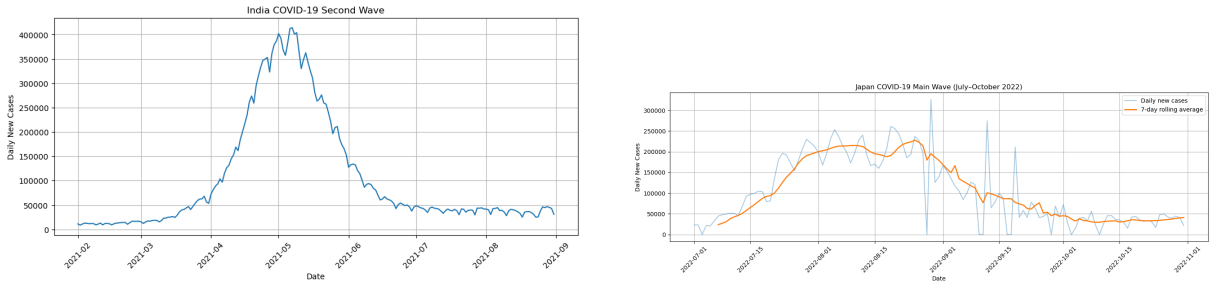


Figure 1: Selected waves. Left: India, daily new cases, February–August 2021. Right: Japan, daily new cases and 7-day average, July–October 2022.

5 Fitting Procedure

5.1 Effective population

Reported cases are far fewer than true infections, and mixing is not uniform. To absorb this, the susceptible pool is treated as a fraction f of the national population, $N_{\text{eff}} = f \cdot N_{\text{pop}}$, and f is fitted along with β and γ. The initial susceptible count is $S_0 = N_{\text{eff}} - I_0 - R_0^{\text{init}}$.

5.2 Objective function

For a trial (β, γ, f) the model is integrated by RK4 with a one-day step. Predicted new infections on day t are $S(t-1) - S(t)$, summed into weeks (Sunday-ending) and compared with observed weekly new cases. The error is

$$\mathcal{E}(\beta, \gamma, f) = \sum_w \left(\frac{\hat{c}_w - c_w}{\max_w c_w} \right)^2, \tag{7}$$

which makes it dimensionless and comparable between countries. Aggregating to weeks removes day-of-week reporting effects.

5.3 Two-stage grid search

A coarse grid with 25 values of $\beta \in [0.05, 1.0]$, 25 of $\gamma \in [0.03, 0.5]$ and 8 values of f between 0.01 and 0.2 is searched exhaustively. A fine grid ($20\times20\times10$ points) is then built around the coarse optimum, with β within ±0.05, γ within ±0.03 and f within $\pm40\%$. Fit quality is reported as weekly RMSE and R^2.

6 Results: Numerical Methods

The test problem uses $N = 1000$, $\beta = 0.3$, $\gamma = 0.1$ (so $R_0 = 3$), $S_0 = 990$, $I_0 = 10$, $R_0 = 0$ on $t \in [0, 160]$, with 40 to 1280 steps.

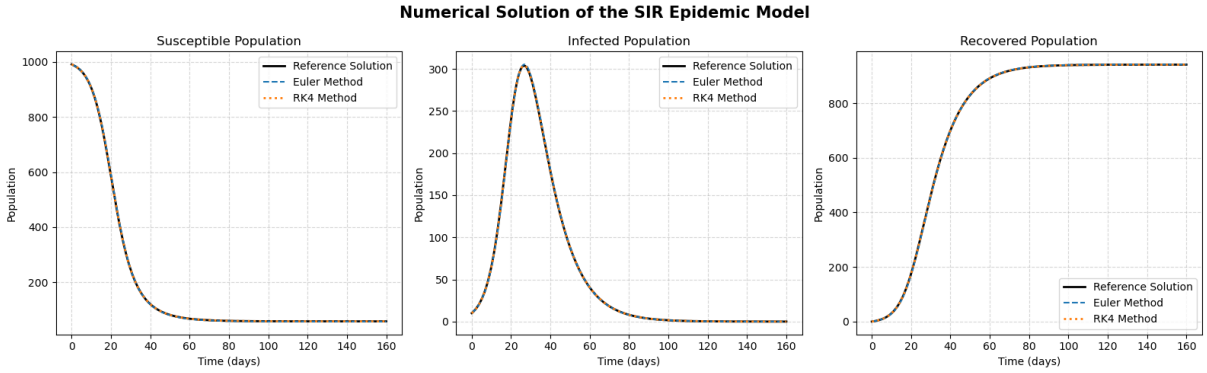


Figure 2: Numerical solutions S, I, R from Euler and RK4 against the reference solution. At this scale the three curves are indistinguishable by eye.

Table 1: Maximum absolute error against the reference solution.

Step size h	Euler error	RK4 error
4.000	1.983×10^{2}	7.092×10^{-1}
2.000	1.029×10^{2}	5.816×10^{-2}
1.000	5.196×10^{1}	4.173×10^{-3}
0.500	2.607×10^{1}	2.796×10^{-4}
0.250	1.304×10^{1}	1.808×10^{-5}
0.125	6.522×10^{0}	1.150×10^{-6}

Halving h roughly halves the Euler error and divides the RK4 error by about 16, as expected. The fitted slopes are 0.988 for Euler and 3.858 for RK4 (Figure 3). The RK4 slope is a little below 4 because the coarsest steps are outside the asymptotic regime. At $h = 1$, RK4 is about 12,000 times more accurate than Euler, and at $h = 0.125$ it is about 5.7×10^{6} times more accurate.

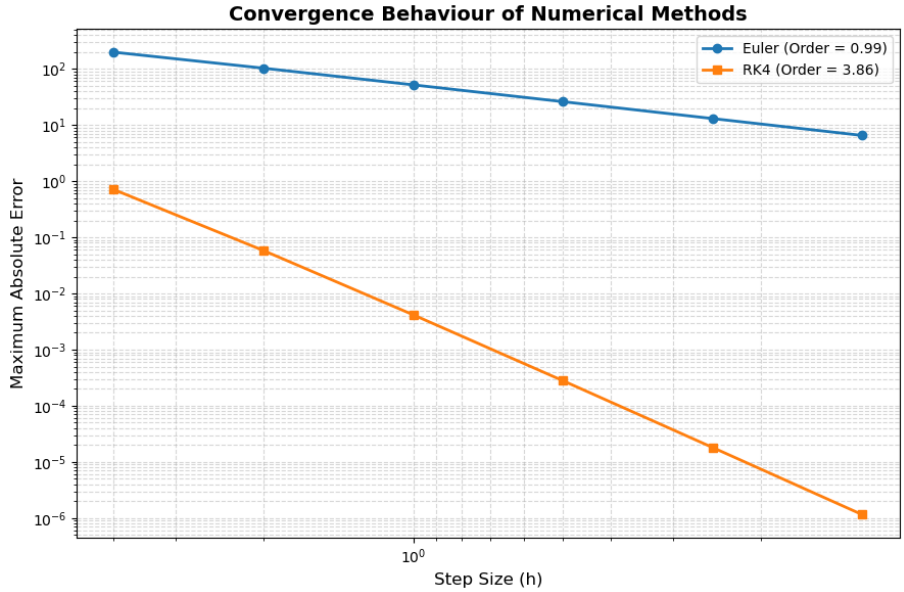


Figure 3: Log–log convergence plot. Observed order: Euler 0.99, RK4 3.86.

6.1 Analytical invariant

For these parameters $C = -1299.235$. Using the finest step ($h = 0.125$), the maximum drift of the invariant is 3.56 for Euler and 6.12×10^{-8} for RK4. Euler's error appears early in the epidemic and then stays constant (Figure 4), whereas RK4's drift is close to rounding level.

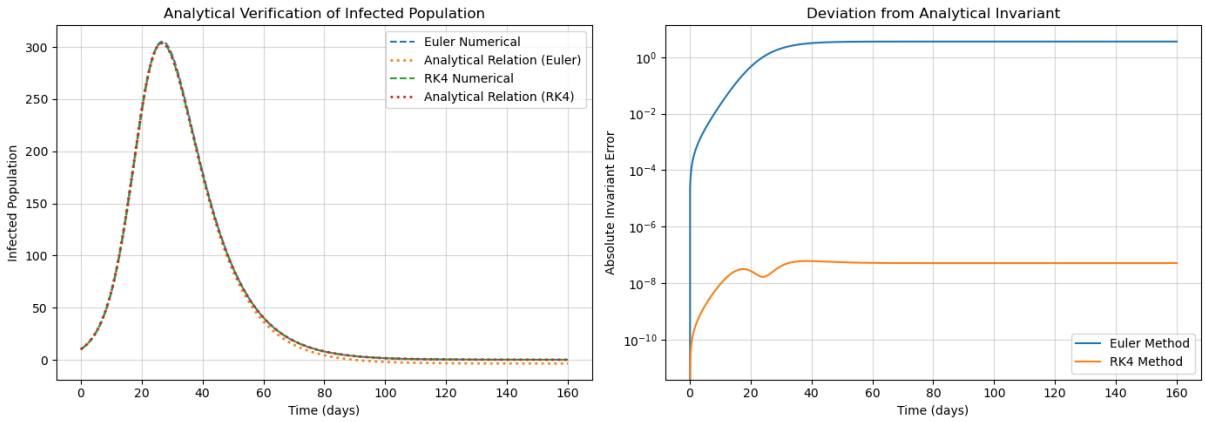


Figure 4: Left: infected population from the numerical $S(t)$ and the analytical relation. Right: absolute deviation from the invariant, on a log scale.

7 Results: Parameter Estimation

7.1 India, second wave

The coarse search gave $\beta = 0.129$, $\gamma = 0.050$, $f = 0.03$ (error 0.656). The fine search gave $\beta = 0.0950$, $\gamma = 0.0196$, $f = 0.026$ (error 0.500). This gives $N_{\text{eff}} = 37.06$ million, 2.6% of the population of 1.425 billion, and an average infectious period of 51 days. The weekly fit has RMSE 347,831 and $R^2 = 0.811$. The model is too high during the rise, too low at the peak, and peaks somewhat later than the data (Figure 5).

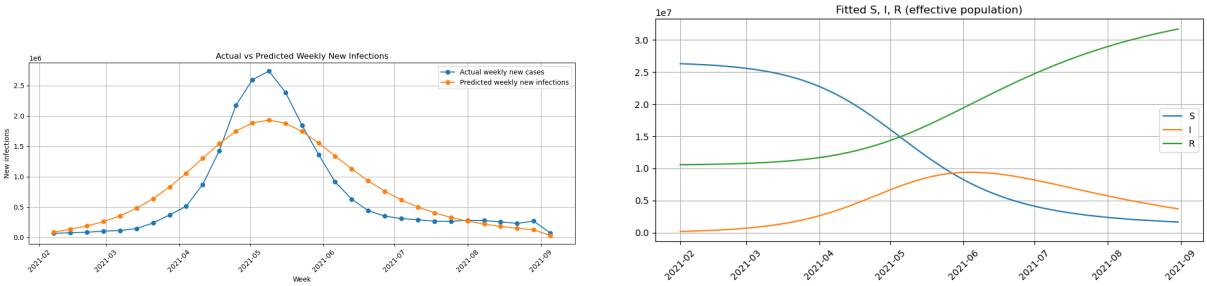


Figure 5: India. Left: actual and predicted weekly new infections. Right: fitted S, I and R for the effective population.

7.2 Japan, summer wave

The coarse search gave $\beta = 0.248$, $\gamma = 0.089$, $f = 0.20$ (error 0.223) and the fine search $\beta = 0.2663$, $\gamma = 0.1093$, $f = 0.2267$ (error 0.112). Hence $N_{\text{eff}} = 28.33$ million (22.7%), with an infectious period of 9.1 days. The fit is closer than for India: RMSE 115,079 and $R^2 = 0.949$ (Figure 6).

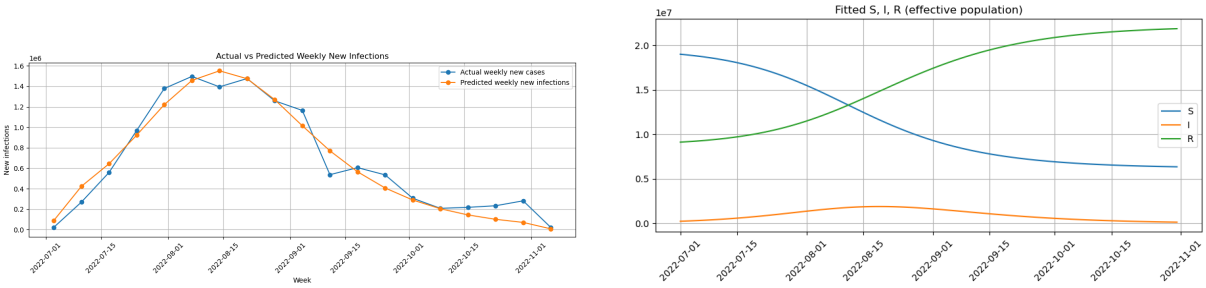


Figure 6: Japan. Left: actual and predicted weekly new infections. Right: fitted S, I and R.

7.3 Comparison

Table 2: Fitted parameters and goodness of fit.

	India	Japan
Study period	Feb–Aug 2021	Jul–Oct 2022
β (per day)	0.095	0.2663
γ (per day)	0.0196	0.1093
$R_0 = \beta/\gamma$	4.85	2.44
Infectious period $1/\gamma$ (days)	51.0	9.1
Effective population fraction	2.6%	22.67%
Effective population	37,061,003	28,332,786
Weekly RMSE	347,831	115,079
Weekly R^2	0.811	0.949

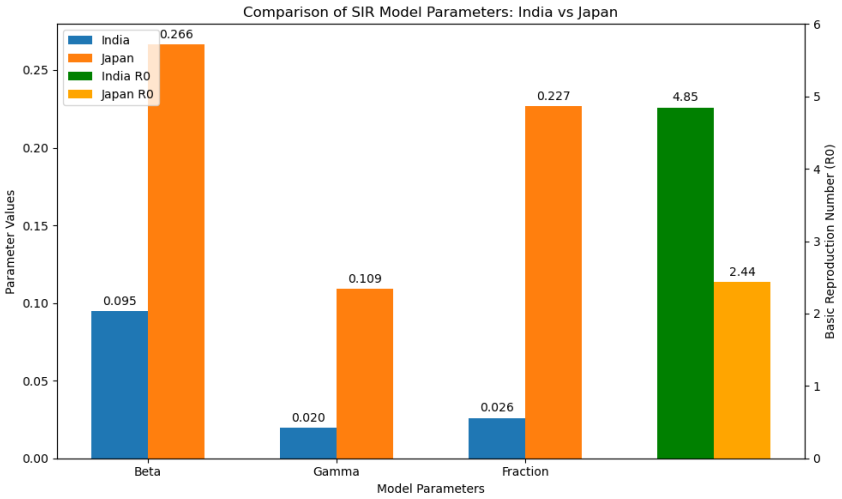


Figure 7: Fitted β, γ, effective fraction (left axis) and R_0 (right axis) for the two countries.

India has the higher R_0 but a very low γ and a small effective population; Japan has a lower R_0, faster turnover and a much larger effective fraction. Because β, γ and f trade off against one another in the fit, these differences should be read as properties of the fitted curves rather than as measured biology (Section 10).

8 Results: Flattening the Curve

In each country β is reduced while γ and the initial state are kept, representing masks, distancing and similar measures. The model is solved by RK4 for 210 days from the fitted initial state. For India β goes from 0.095 to 0.070 (-26%) and for Japan from 0.2663 to 0.1951 (-27%). The healthcare capacity lines are illustrative thresholds, set at 3×10^6 for India and 7.5×10^5 for Japan.

Table 3: Scenario results. R_0 in the India rows uses $\gamma = 0.019$, as in the scenario code.

Country	Scenario	R_0	Peak infections	Peak day	Peak reduction
India	Higher β (0.095)	4.85	9,749,217	123	–
	Lower β (0.070)	3.68	6,789,588	175	30%, delay 52 d
Japan	Higher β (0.2663)	2.44	1,890,494	49	–
	Lower β (0.1951)	1.78	505,059	65	73%, delay 16 d

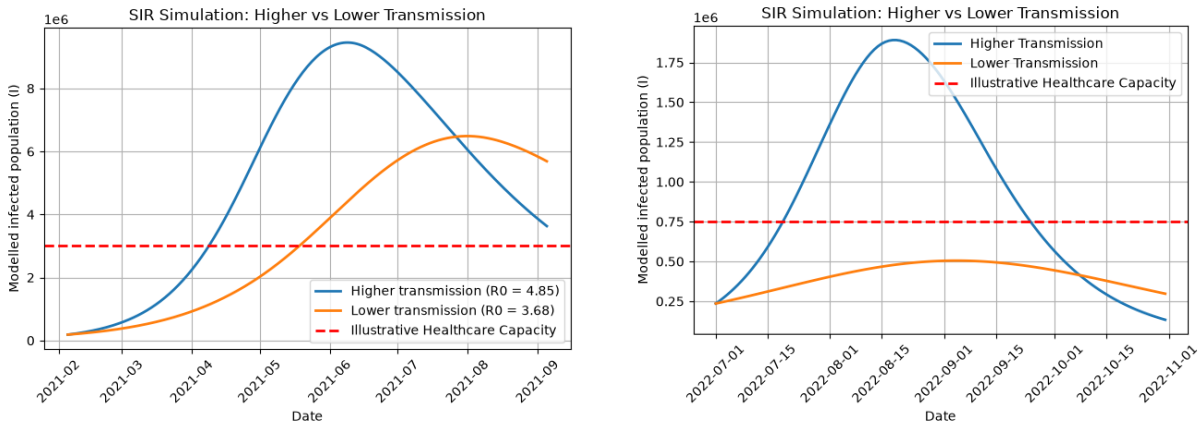


Figure 8: Modelled infected population under higher and lower transmission. Left: India. Right: Japan. Dashed red line: illustrative healthcare capacity.

A similar relative cut in β has very different effects. In India the lower-β epidemic is delayed by 52 days but the peak remains high and exceeds the capacity line. In Japan the same cut lowers the peak by almost three quarters, to below the capacity line, and delays it by 16 days. The reason is that the final peak size depends on how far R_0 lies above one. With R_0 reduced from 2.44 to 1.78 in Japan the epidemic is much closer to the threshold, whereas India's R_0 remains well above one (3.68) after the reduction.

9 Historical Bombay Plague Data Application

The same framework is applied to weekly reported plague cases in Bombay City. The first wave, 40 weeks from 25 September 1905 to 30 June 1906, is used with $I_0 = 37$, $R_0^{\text{init}} = 0$ and the 1891 population $N_{\text{pop}} = 821{,}000$. Since the data are weekly, β and γ are expressed per week and RK4 uses a step of one week. The two-stage grid search of Section 5 (coarse: $\beta \in [0.1, 4]$, $\gamma \in [0.1, 3]$, $f \in [0.05, 1]$; fine: ± 0.4 around the coarse optimum) gives $\beta = 1.107$, $\gamma = 0.893$ and $f = 0.0464$ (error 0.268, against 1.160 for the coarse grid). Hence $N_{\text{eff}} = 38{,}118$ (4.64% of the population), $R_0 = 1.24$ and an infectious period of $1/\gamma = 1.12$ weeks, a short course consistent with plague. The weekly fit has RMSE 70.2 cases and $R^2 = 0.923$ (Figure 9). The model reproduces the rise

and decay of the wave but, being smooth and symmetric, peaks at about 700 cases against an observed maximum near 850, and overestimates the early and late tails.

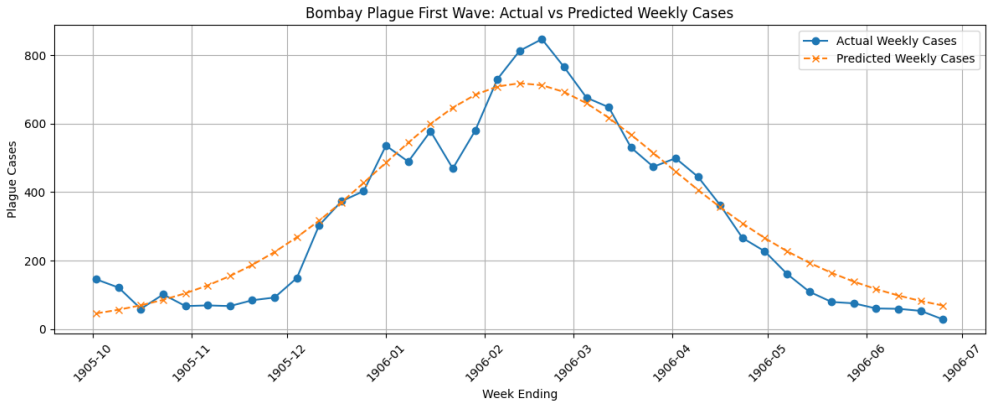


Figure 9: Bombay plague, first wave (1905–06): actual and RK4-predicted weekly new cases.

As a scenario test, $\beta = 1.95$ with $\gamma = 1.70$ ($R_0 = 1.15$) gives a modelled peak of 7,106 infected in week 27, whereas reducing β to 1.55 ($R_0 = 0.91$) stops the outbreak at once, leaving the peak at $I_0 = 37$. A supplementary $A\,\mathrm{sech}^2(wt - \phi)$ curve fitted to the first 35 weekly death records gives $A = 585.6$, $w = 0.1145$, $\phi = 2.446$, lower than the literature values 890, 0.2 and 3.4.

10 Discussion

10.1 Choice of solver

For smooth, non-stiff problems like SIR, RK4 is clearly the better choice. For the same step, it is thousands of times more accurate than Euler and keeps the conserved quantity essentially exact. Euler needs steps hundreds of times smaller to reach comparable accuracy. For the COVID fits, a one-day RK4 step is more than adequate, since the rates are much smaller than one per day.

10.2 Quality of the fits

The Japan fit is good ($R^2 = 0.95$) and its $R_0 = 2.44$ and 9-day infectious period are in a plausible range for an Omicron-era wave. The India fit is weaker: it overestimates the early rise and underestimates the peak, because a constant-parameter SIR curve is symmetric whereas the real wave rose sharply and fell quickly.

10.3 Identifiability and boundary warnings

The notebook flags two boundary hits. For India, $\gamma = 0.0196$ is the lower edge of the fine γ grid (0.0196 to 0.0796), so the true optimum may lie at an even lower γ, and the 51-day infectious period is probably an artefact: clinical infectious periods are one to two weeks. For Japan, the coarse f optimum of 0.20 was at the upper edge of the coarse grid, so the fraction could be higher still, though the fine grid (0.12–0.28) then selected an interior value of 0.2267. A low γ together with small f is a way for the model to produce a wave of the right size and timing from limited data, which shows that the parameters are only weakly identified by a single wave of new-case counts.

10.4 Limitations

- Constant β and γ: real interventions, vaccination and variants change transmission over time.
- Homogeneous mixing: no age structure, geography or networks.
- The 14-day infectious period used to build the initial conditions is an assumption and differs from the fitted $1/\gamma$.
- f is a catch-all for under-reporting and incomplete susceptibility.
- Reporting artefacts, such as the days with zero reported cases in the Japan data, are not modelled.
- The healthcare capacities are illustrative.

11 Conclusion and Future Work

RK4 reaches its theoretical fourth-order accuracy (3.86 observed) and preserves the SIR invariant to 6×10^{-8}, whereas Euler is first-order (0.99) with an invariant drift of 3.6. A grid search on weekly new cases fits the Japan summer wave well ($R^2 = 0.95$, $R_0 = 2.44$) and the India second wave less well ($R^2 = 0.81$, $R_0 = 4.85$). A roughly 26% reduction in β lowers the modelled peak by 30% in India and 73% in Japan, which shows that early interventions matter most when R_0 is closer to one.

Natural extensions are time-varying $\beta(t)$, SEIR or SEAIR compartments, vaccination, stochastic versions, and Bayesian estimation with uncertainty intervals, which would address the identifiability problems noted above.

References

[1] W. O. Kermack and A. G. McKendrick, "A contribution to the mathematical theory of epidemics," *Proc. R. Soc. A*, 115 (1927), 700–721.

[2] Our World in Data, COVID-19 dataset (based on Johns Hopkins University CSSE).

[3] J. R. Dormand and P. J. Prince, "A family of embedded Runge–Kutta formulae," *J. Comput. Appl. Math.*, 6 (1980), 19–26.

[4] O. Diekmann, J. A. P. Heesterbeek and J. A. J. Metz, "On the definition and the computation of the basic reproduction ratio," *J. Math. Biol.*, 28 (1990), 365–382.